Session-IX: Crop adoption survey results: November last week

Program schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	20 min.	Recap.	Large group discussion	Charts and markers.	Participants recollect the contents discussed in eighth session.
2	45 min.	Observation of LTE and data collection.	Small group discussion	FWS group kit, charts and markers.	Participants observe the changes in experimental and control plots and analyze the data.
3	30 min.	Sharing crop adoption Survey Results.	Large group discussion	Charts, markers, banner on crop plans.	Participants analyze the crop adoption survey results.
4	30 min.	Estimation of actual water requirement for the HU.	Large group discussion	Charts, markers, banner on crop plans.	Participants estimate the actual water requirement for the HU.
5	30 min.	Group dynamics: Data accuracy.	Game		Participants understand the importance of data accuracy.
6	20 min.	Short Study: Water holding capacity.	Demonstration	Three types of soils, water bottles, plastic trays, thread, water and organic manure.	Participants discuss impact of organic fertilizers in increasing water holding capacity of the soils.
7	5 min.	Homework: Existing alternate irrigation techniques.	Large group		Participants collect data on existing alternate irrigation techniques in their respective village.
8	10 min.	Evaluation.	Large group discussion	Charts and markers.	Participants share their opinions on the conduct of FWS sessions and methods used in the session.

Learning objectives

By the end of the session participants will be able to:

- Recap the previous session,
- Observe LTE and analyze data,
- Share crop adoption survey results,
- Estimate the actual water requirement for the HU, and
- Understand the importance of collective decision making.

Process

1. Recap Time: 20 min.

Objective : Participants recap the contents discussed in the previous session.

Content : Session eight contents

- Design of Long Term Experiment.
- Selection of Collaborator.

Method : Large group discussion

Materials : Charts and markers.

Process

- Welcome participants to the ninth session.
- Start the session with FWS pledge.
- In the large group, encourage each participant to share a content that was discussed in previous session and the learning from that one. Use pictures or banners that were used in the session.
- While facilitating the recap, write down the responses on a chart paper.
- Discuss the contents and the learning from the previous session.

2. Observation of LTE and data collection Time: 45min.

Objective : Participants observe and Long Term Experiment (LTE) and collect data.

Content :

- Observation of LTE, and
- Data collection.

Method : Small group discussion

Materials : FWS group kit, charts and markers.

Process

- Enquire whether they were successful in setting up the LTE and ask whether we need to observe the LTE.
- In large group ask the participants to list the features that need to be observed. (Remind them of the list of observation items that were discussed in the previous session).
- Inform the participants that they would need work in their respective learning groups and note observations.
- Enquire if they have the necessary tools to do the observations. Discuss the usage of each tool.
- Encourage the groups to observe the items based on the type of the experiment and assist them in the process of data collection.
- Ask the groups to analyze the observations and present to the large group.
- Sum up the discussion.

3. Sharing crop adoption survey results Time: 30 min.

Objective : Participants analyze the crop adoption survey results.

Content : Crop adoption survey results.

Method : Large group discussion

Materials : Charts, markers and banner on crop plans.

Process

- Ask participants from each habitation to share their crop adoption survey details. Use the following table to summarize the data for the entire HU.

Village	Paddy (Acres)	Groundnut (Acres)	Chilly (Acres)	Black gram (Acres)	Sweet orange (Acres)	Total (Acres)
Abc						
Bcd						

- Present the banner that displayed crop plans shared during the CWB exercise.
- Compare the above output with the information in the banner and discuss the changes in crops and area.

4. Estimation of actual water requirement for the HU Time: 30 min.

Objective : Participants estimate the actual water requirement for the HU.

Content : Actual water requirement.

Method : Large group discussion

Materials : Charts, markers and banner on crop plans.

Process

- Based on the outcome of the crop adoption survey results, share with the participants about the increase or decrease in area of the crop.
- In case there is an increase/ decrease in crop area, estimate the water requirement. This will be the actual water requirement for Rabi season.
- Based on the actual draft, share the change in groundwater balance by the end of May.

5. Group dynamics: Data accuracy Time: 30 min.

Objective : Participants understand the importance of data accuracy.

Content : Data accuracy.

Method : Demonstration

Materials : Three transparent plastic glasses and color.

Process

- Ask participants to place three plastic transparent glasses on a table so that all participants can observe the process.
- Ask one participant to pour some water into first glass and some water into second glass.
- Next, ask one more participant to pour some water into third glass. But the water turns into different color.
- Encourage participants to discuss about color of water if we mix all three glasses of water into one glass.
- Summarize the discussion by sharing the following words. When we use the data to estimate groundwater recharge or draft, the data should be accurate. Even if one

parameter like rainfall received or pumping days or average draft is wrong, entire calculation become wrong and gives a wrong value. So, we need to take care during data collection.

6. Short Study: Water holding capacity Time: 20 min**.**

Objective : Participants will be aware of the impact of organic fertilizers in increasing water holding capacity of the soils.

Rationale : Farmers use chemical fertilizers in large quantities the use of which changes the soil texture and decrease the water holding capacity and soil fertility. Demonstration of the use of organic fertilizers in improving the water holding capacity of the soil will help participants understand better.

Here below are two alternate ways of demonstrating this.

Model One : Use of soils in bottles and water

Materials : Transparent plastic bottles (3 No's), soil, vermi-compost, water, three pieces of cloth, physical balance, 250 grams measuring weight, 1 litre bottle, three trays and farm yard manure (FYM)

Method of Preparation:

- Take three transparent plastic bottles
- Take same type of soil and divide it into three parts
- Put 500 gm. of soil in first bottle, mix well 250 gm of soil and 250 gm vermi compost and put the mixture in second bottle. Mix well 250 gm of soil and 250 gm FYM and put this mixture in third bottle.

Process

- Pour 1 liter water in each bottle
- Cover the top of each bottle with thin cloth and tie it and keep it for 5 minutes
- Keep each bottle upside down in a tray
- Measure the water collected in each tray.
- More water will be collected from the bottle which is filled with only soil.

- From this experiment it can be demonstrated that the soils which are mixed with organic fertilizers have more water holding capacity.

Model two : Use of three types of soils and one organic manure

Materials : 6 water bottles, 6 plastic trays, cotton cloth, three types soil (each 500 grams), thread, water and organic manure

Method of Preparation:

- Remove the lower base portion of all water bottles
- Close the mouth portion of the bottle with cotton cloth tightly
- Fill three bottles with three different soils
- Fill the other three bottles with soil mixed with organic manure (3 parts soil + 1 part organic manure)
- Hang the filled bottles upside down using the thread.
- Keep a tray under each plastic bottle

Usage:

- Pour same quantity of water slowly on top of the soil in each bottle simultaneously and keep them for some time say, five minutes.
- Measure the water collected in the trays
- Discuss the difference in quantity of water collected from three bottles filled with three different soils,
- Next, discuss the difference in quantity of water collected from the three bottles filled with soil and organic manure.
- Discuss the water holding capacity of different soils and the impact of organic manure in increasing water holding capacity.

7. Homework: Irrigation practices in the village Time: 5 min**.**

Request participants to collect the data on existing alternate irrigation techniques practiced by other farmers in the village.

8. Evaluation Time: 10 min**.**

- Ask participants the following questions to evaluate the day's learning. Pose the questions to the large group:

:

- Why do we need crop adoption details?
- Why do we estimate the actual water requirement?

Encourage participants to share their learning and thank the participants for their inputs and participation in the session.